

DO AUDIO DISTANCE FUNCTIONS HEAR TIMBRE MODULATION? A MACHINE LISTENING STUDY

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ABSTRACT

Audio distance functions are widely used in deep inverse audio problems like sound matching, yet their alignment with human perception of dynamic timbre modulation remains uncharacterized. In light of this, we conduct a MUSHRA-style listening test with 54 participants evaluating human sensitivity to modulation amplitude, frequency, and irregularity across warmth, brightness, and richness timbre qualities using wavetable synthesis. Listeners demonstrate robust perceptual invariance to underlying timbre and wavetable source, with perceived difference scaling linearly with modulation amount. We then evaluate nine audio distance functions across STFT-based, wavelet-based, and neural representations using linear regression, variance decomposition, and correlation with human ratings. Wavelet scattering transforms provide the most consistent alignment with human perception across modulation types, while STFT-based methods fail to resolve modulation frequency, and neural methods exhibit fragmented, embedding dependent competencies. Our findings offer empirical evidence to guide the principled selection and design of perceptually grounded loss functions for deep inverse problems involving modulated audio. We make our code, audio samples, and listening test data available.

Index Terms— modulation perception, listening study, audio distance functions, perceptual loss, wavelet scattering transforms

1. INTRODUCTION

Modulation is fundamental to audio production, enabling sound to evolve over time through dynamic variation of synthesis parameters [1, 2, 3]. Amplitude, frequency, filter, and wavetable modulation are widely used to shape timbre and introduce movement. Capturing these dynamic behaviors is essential for deep inverse audio problems, such as neural synthesis, synthesizer programming, and sound matching of time-varying audio systems [4, 5, 6]. However, while audio distance / loss functions are routinely used to train models and evaluate synthesis fidelity, their alignment with human perception of modulated audio remains poorly understood.

More broadly, evaluating the perceptual alignment of algorithmic audio distances is an active research direction, motivated by the failure of spectral losses to reliably capture fundamental auditory attributes such as pitch [7] or generalize across distinct musical timbre spaces [8]. While learned perceptual metrics [9, 10] and neural embeddings offer alternatives, their alignment with human ratings remains inconsistent and heavily model-dependent [11, 12]. Alternatively, wavelet transforms like time scattering [13] and joint time–frequency scattering (JTFS) [14, 15] provide representations grounded in auditory neuroscience, modeling spectrotemporal receptive fields in the mammalian auditory cortex [16, 17]. Yet, de-

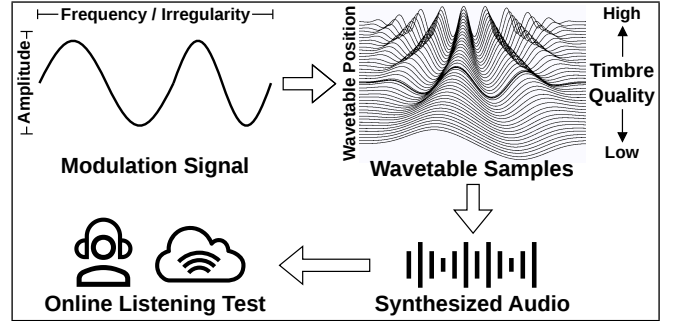


Fig. 1. A modulation signal can vary in amplitude, frequency, and irregularity to scan through a wavetable that spans a range of timbre qualities. The resulting audio stimulus was rated by listeners for perceived difference from a minimally modulated reference.

spite their capacity to capture spectrotemporal dynamics, existing perceptual evaluations assess only broad, holistic similarity without isolating the rate, depth, or regularity of change. A primary cause is the scarcity of controlled, psychoacoustically grounded listening test data, leaving distance functions uncharacterized across specific timbral modulation types.

To address this gap, we conduct a MUSHRA-style [18] listening test characterizing human perception of timbral modulation and evaluate nine distance functions across three paradigms: Short-Time Fourier Transform (STFT)-based methods, wavelet scattering transforms, and neural audio embeddings. While prior work has investigated amplitude [19, 20] and frequency [19, 21] modulation or static timbre qualities like warmth, brightness, and richness [22, 23], timbral modulation introduces interacting temporal factors such as modulation depth (amplitude), rate (frequency), and periodicity (irregularity). Using wavetable synthesis with pitch and loudness held relatively constant, we systematically isolate these three modulation types and timbre qualities to benchmark perceptual alignment of distance functions against human ratings.

Our contributions are the following:

- We conduct a MUSHRA-style listening test (90 stimuli, 54 participants) and release the resulting perceptual dataset characterizing human sensitivity to modulation amplitude, frequency, and irregularity for warmth, brightness, and richness timbre qualities.
- We find that listener perception is highly invariant to sound source (synthetic or real) and timbre qualities; differences are driven primarily by modulation amount, with salience ranking highest for frequency, then amplitude, and lowest for irregularity.
- We evaluate nine audio distance functions via linear fit, variance decomposition, and correlation analyses, finding that wavelet scattering transforms align most closely with human perception of timbral modulation.

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2. HUMAN PERCEPTION STUDY

2.1. Stimuli

90 stimuli (2 wavetable sources [synthetic vs. real] $\times$ 3 timbre qualities $\times$ 3 modulation types $\times$ 5 modulation amounts) were synthesized using a four-second long modulation signal at 44.1 kHz that scans a wavetable across multiple positions, resulting in a time-varying change in timbre while loudness and pitch remain relatively constant.¹ Six wavetables were chosen, one synthetically generated and one real acoustic source per timbre quality (warmth, brightness, richness), each sweeping linearly along its quality axis (see Fig. 1). For each wavetable, three modulation types drove the wavetable position scan: amplitude (0.1–0.9), frequency (0.25–4.0 Hz), and irregularity (0–50%, i.e. aperiodicity), each tested at five levels with the other two dimensions held fixed. The lowest level served as reference: for amplitude and frequency this retained slight modulation (0.1 depth, 0.25 Hz); for irregularity it was fully periodic (0%).

2.2. Participants

The online listening test followed the MUSHRA standard [18] and consisted of 18 trials. Each trial presented five stimuli from a wavetable source, timbre quality, and modulation type combination, rated for perceived difference from the reference on a 0 (“identical”) to 100 (“completely different”) scale. 54 participants completed the test. Three were removed for using low quality playback devices, and 23 more were removed based on the MUSHRA unreliable rater definition (i.e., the hidden reference is rated >10 for $>15\%$ of trials). Nine additional individual trials were removed where completion took under 24 seconds or all five stimuli received identical ratings, resulting in 28 remaining participants of which 23 have complete data across all 18 trials. The listening test complies with and was approved by the Queen Mary Research Ethics Committee.¹

2.3. Statistical Analysis

A sphericity-corrected four-way repeated-measures ANOVA on the post-processed listening test data ($N = 23$) with the reference ratings removed indicates that all four main effects are significant:

- Wavetable source: $F(1, 22) = 8.84, p < 0.01, \eta_G^2 = 0.00414$
- Timbre quality: $F(2, 44) = 20.5, p < 0.001, \eta_G^2 = 0.0540$
- Modulation type: $F(2, 44) = 75.0, p < 0.001, \eta_G^2 = 0.261$
- Modulation amount: $F(3, 66) = 389, p < 0.001, \eta_G^2 = 0.560$

Bonferroni-corrected paired t -tests confirm that all timbre quality, modulation type, and amount combinations are significant with $p < 0.05$ for warmth vs. richness and $p < 0.001$ for all remaining pairs. Among all two-, three-, and four-way interactions, only two are significant: modulation type $\times$ amount ($p < 0.001$) and wavetable source $\times$ timbre quality ($p < 0.01$). The former is driven by differing perceptual growth rates: while all modulation types scale monotonically, frequency scales substantially steeper than amplitude or irregularity, diverging most at higher amounts. The latter is isolated entirely to warmth, where synthetic wavetables were rated significantly lower than real sources ($\Delta = -8.07, p < 0.001$), whereas brightness and richness exhibited no source differences.

We observe that the vast majority of the variance in the perceptual data (η_G^2) is attributed to modulation type and amount. Accordingly, we visualize the listening test results averaged across wavetable source and timbre quality in the top row of Fig. 2. Perceived difference follows a strong linear trend as modulation amount

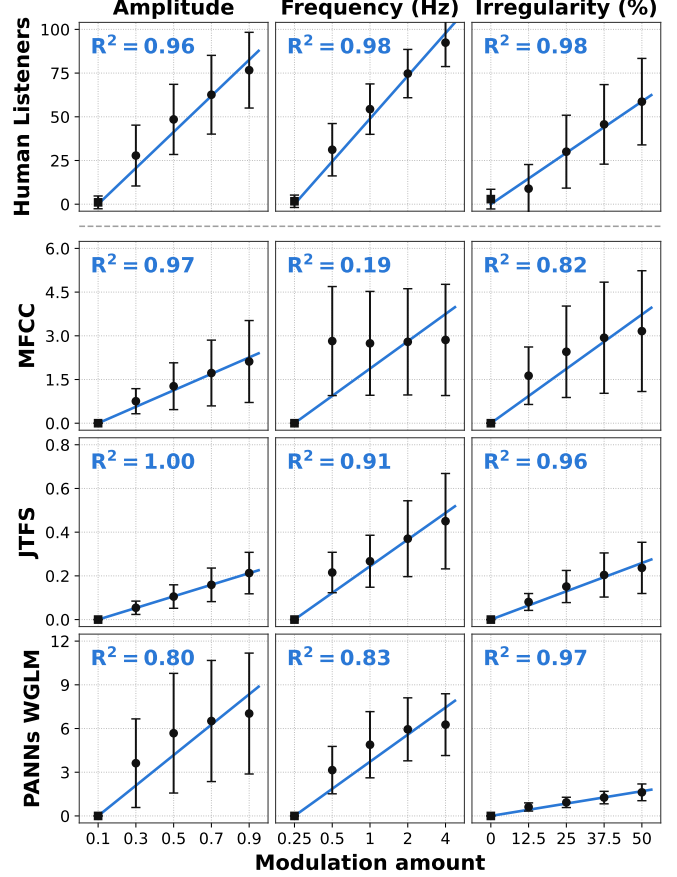


Fig. 2. Perceptual and algorithmic distances between the minimally modulated reference (■) and stimuli with increasing modulation amounts (●) for amplitude, frequency, and irregularity. Markers and error bars denote mean and standard deviation, respectively. Solid blue lines are zero-anchored linear fits with annotated R^2 . The top row shows human listening test ratings (perceptual benchmark), followed by the highest average R^2 scoring method from each distance family: STFT-based (MFCC), wavelet-based (JTFS), and neural (PANNs WGLM). All R^2 values are in Table 1 and all plots for methods not shown here are available at the companion website.¹

increases for all three modulation types ($R^2 \geq 0.96$), however, perceptual salience differs: at the highest level ratings reached ~ 75 for amplitude and ~ 90 for frequency, but only ~ 60 for irregularity. Additional statistical analysis of the listening test data can be found in our prior work [24].

3. AUDIO DISTANCE FUNCTION EVALUATION

We conduct three analyses to determine which audio distance / loss functions for deep inverse audio problems are most aligned with human perception of timbral modulation: linear regression (Sec. 3.1), variance decomposition (Sec. 3.2), and monotonic / linear correlation analysis (Sec. 3.3). We compare three STFT based losses: log + linear multiscale spectral (MSS) loss [25], a SOTA hyperparameter-tuned revisited MSS loss [26], and Mel-frequency cepstral coefficients (MFCC) L_1 distance [27]; the Euclidean distance of two wavelet scattering transform coefficients: time scattering (Scat1D), and JTFS; and the Euclidean distance of four neural audio embeddings: VGGish [28], EnCodec 48 kHz [29], MS-CLAP [30], and PANNs Wavegram Logmel (WGLM) [31].

¹ Complete stimuli generation details, timbre quality definitions, code, audio samples, plots, ethics approval, and listening test data provided at: https://christhetree.github.io/mod_perception

Table 1. Zero-anchored linear fit R^2 values and distance family averages across amplitude, frequency, and irregularity modulation types. Human listening test data acts as the perceptual benchmark. Plots are provided for underlined methods in Fig. 2 which are the highest average R^2 scoring method from each distance family.

Method	Amp. R^2	Freq. R^2	Irreg. R^2
<u>Human Listeners</u>	0.96	0.98	0.98
MSS Log + Linear	0.78	0.32	0.66
MSS Revisited	0.95	0.13	0.76
<u>MFCC</u>	0.97	0.19	0.82
Scat1D	1.0	0.89	0.96
<u>JTFS</u>	1.0	0.91	0.96
VGGish	0.84	0.92	0.82
EnCodec 48 kHz	0.98	0.22	0.74
MS-CLAP	0.94	0.99	0.45
<u>PANNs WGLM</u>	0.80	0.83	0.97
STFT-based mean	0.91	0.21	0.75
Wavelet-based mean	1.0	0.90	0.96
Neural mean	0.91	0.84	0.76

The STFT and neural methods are widely adopted loss functions or evaluation distances for sound matching tasks. Although less conventional, wavelet scattering transforms like Scat1D and JTFS (see Sec. 1) provide the temporal invariance and multi-resolution analysis required to sound-match new classes of synthesizers and modulated audio [32, 33], motivating their inclusion. Prior work has also found that JTFS, MS-CLAP, and PANNs WGLM correlate with human perception of music and audio samples [11, 12, 34, 35].

For each method and listening test trial, we compute the distance between the reference and modulated stimuli, resulting in nine sets of 72 numerical values which are used in the following three analyses. These distances averaged across wavetable source and timbre quality are plotted for MFCC, JTFS, and PANNs WGLM in Fig. 2 alongside the human listening test data. All distance functions are differentiable, deterministic, non-negative ($d(x, y) \geq 0$), symmetric ($d(x, y) = d(y, x)$), and reflexive ($d(x, x) = 0$).

3.1. Linear Regression

Given the strong linear relationship between perceived difference and modulation amount across all three modulation types (see Sec. 2.3 and Fig. 2, top row), the first analysis evaluates whether the nine distance functions exhibit comparable linearity by fitting a linear model to each. To account for the reflexivity of the distance functions and the MUSHRA standard’s unreliable rater definition (see Sec. 2.2), linear regression lines are anchored at $d_{\text{method}}(x_{\text{ref}}, x_{\text{ref}}) = 0$ for the reference stimulus. The resulting R^2 values are summarized in Table 1 and visualized in Fig. 2.

From Table 1 and Fig. 2, we see that wavelet-based distance functions match the linearity of the human perceptual data most faithfully, achieving the closest mean R^2 values across all three modulation types. In stark contrast, all STFT-based distance functions experience a severe breakdown for the frequency category, yielding poor linearity despite performing adequately on amplitude and irregularity. We believe this failure stems from the fixed time-frequency tiling of the STFT, which smears rate changes across spectral bins rather than resolving modulation frequency. The EnCodec 48 kHz embedding exhibits a similar failure for frequency which we attribute to its temporal chunking mechanism. However, MS-CLAP has near-perfect linearity for frequency ($R^2 = 0.99$), but

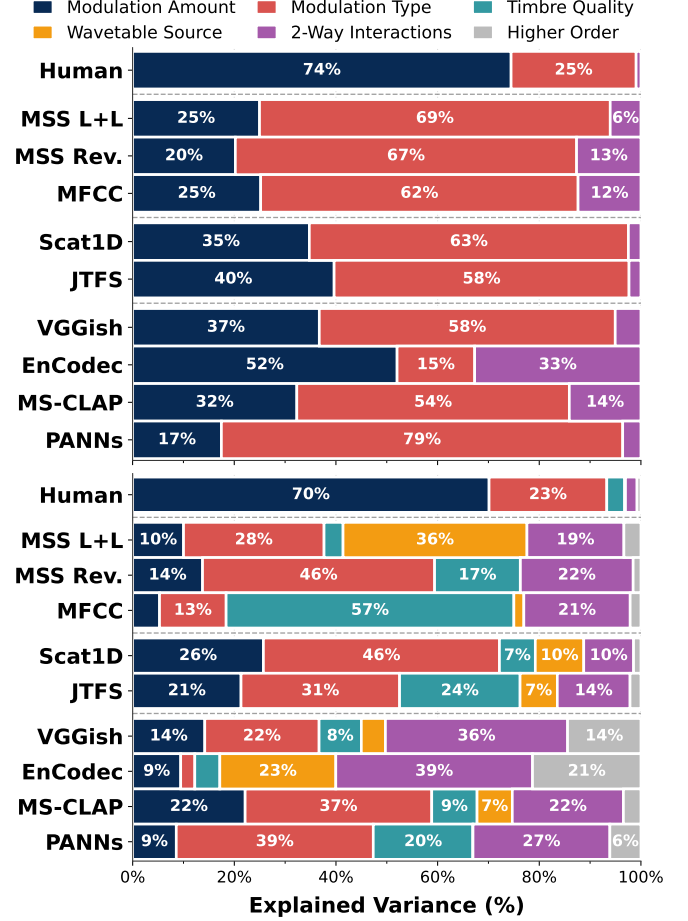


Fig. 3. Interaction-pooled ANOVA variance decompositions comparing human perceptual ratings with audio distance functions.

Top: marginalized across wavetable source and timbre quality given listeners’ near-invariance to both (modulation type $\times$ amount as the two-way interaction). **Bottom:** full decomposition across all four factors with all two-way interactions combined into one term.

collapses for irregularity ($R^2 = 0.45$). Of the neural methods, only PANNs WGLM and VGGish maintain balanced linearity across all three modulation types.

3.2. Variance Decomposition

The second analysis decomposes the variance of the listening test ratings and computed audio distances across wavetable source, timbre quality, modulation type, and amount, as well as two-way and higher-order interactions. Based on the analysis in Sec. 2.3, a perceptually aligned distance function should attribute most of its variance to modulation type and amount, with negligible contributions from other factors and interactions.

Since the distance functions are deterministic algorithms evaluated across an unreplicated orthogonal grid ($N = 72$), they have no subject-level random variance, unlike the listening test data. We therefore use an unreplicated, interaction-pooled, factorial ANOVA purely descriptively to partition variance without inferential hypothesis testing for each distance function and the participant-averaged ratings. Given listeners’ near-invariance to wavetable source and timbre quality, the top of Fig. 3 presents a marginalized decomposition isolating modulation type, amount, and their specific two-way interaction (modulation type $\times$ amount). The full four-factor vari-

Table 2. Audio distance function correlation coefficients and distance family averages for the human perceptual data. Noise ceilings are reported as split-half bootstrapped 95% confidence intervals. Highest correlation values are in **bold**; second highest are underlined. Statistical significance is denoted by asterisks (unstarred: $p \geq 0.05$ or N/A for the last three aggregate rows, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Method	Amplitude		Frequency		Irregularity	
	Spearman (ρ) $\uparrow$	Pearson (r) $\uparrow$	Spearman (ρ) $\uparrow$	Pearson (r) $\uparrow$	Spearman (ρ) $\uparrow$	Pearson (r) $\uparrow$
Noise Ceiling	[0.940, 0.989]	[0.947, 0.989]	[0.974, 0.991]	[0.983, 0.995]	[0.947, 0.989]	[0.952, 0.990]
MSS Log + Linear	0.570 **	0.537 **	0.217	0.196	0.577 **	0.514 *
MSS Revisited	0.813 ***	<u>0.823</u> ***	0.145	0.0595	0.894 ***	0.855 ***
MFCC	0.653 ***	0.667 ***	0.169	0.119	0.600 **	0.576 **
Scat1D	0.817 ***	<u>0.823</u> ***	<u>0.766</u> ***	<u>0.743</u> ***	<u>0.879</u> ***	0.887 ***
JTFS	<u>0.816</u> ***	0.826 ***	0.642 ***	0.614 **	0.779 ***	0.787 ***
VGGish	0.452 *	0.363	0.704 ***	0.656 ***	0.703 ***	0.768 ***
EnCodec 48 kHz	0.806 ***	0.789 ***	0.116	0.0776	0.415 *	0.373
MS-CLAP	0.763 ***	0.757 ***	0.870 ***	0.854 ***	0.395	0.408 *
PANNs WGLM	0.560 **	0.562 **	0.622 **	0.610 **	0.861 ***	<u>0.861</u> ***
STFT-based mean	<u>0.679</u>	<u>0.676</u>	0.177	0.125	<u>0.690</u>	<u>0.648</u>
Wavelet-based mean	0.816	0.825	0.704	0.678	0.829	0.837
Neural mean	0.645	0.618	<u>0.578</u>	<u>0.549</u>	0.593	0.603

ance decomposition is shown in the bottom of Fig. 3, with all two-way interactions combined into a single term.

The variance decomposition in Fig. 3 reveals a fundamental divergence between human listeners and audio distances. In the perceptual data, modulation amount accounts for the overwhelming majority of variance (74% marginalized, 70% full), followed by modulation type (25% marginalized, 23% full). Crucially, human listeners remain almost entirely invariant to timbre quality, wavetable source, and higher-order interactions. None of the distance functions replicate this and instead invert this hierarchy: modulation type explains substantially more variance than modulation amount for nearly all methods. Furthermore, several are dominated by factors human listeners ignored: MFCC is mostly driven by timbre quality, MSS Log + Linear is dominated by wavetable source, and EnCodec 48 kHz suffers from severe interaction artifacts. Overall, wavelet scattering transforms and MS-CLAP best isolate modulation type and amount while suppressing spurious interactions, with Scat1D coming closest to the human data in the full variance decomposition. JTFS performs slightly worse and appears to be more sensitive to timbre quality and higher order interactions than Scat1D, which is consistent with its explicit joint filtering across the log-frequency axis.

3.3. Monotonic and Linear Correlation

The last analysis measures the correlation between audio distance functions and the human perceptual ratings. Following our variance decomposition findings (Sec. 3.2), we assess correlation separately for each modulation type (six trials, 24 stimuli each). To establish an interpretable benchmark, we estimate the human noise ceiling via 1,000 split-half bootstrap iterations, defining the empirical upper bound imposed by inter-listener variability. Perceptually aligned distance functions should have positive correlation coefficients approaching this ceiling.

We compute group-level (participant-averaged) Spearman’s rank (ρ) and Pearson’s (r) correlation coefficients for each method. Spearman’s ρ measures monotonic concordance (preserving perceptual rank ordering invariant to non-linear compression), while Pearson’s r quantifies linear interval alignment (proportionality to human rating magnitudes). When used as loss functions for neural network training, monotonic alignment is arguably more critical than linearity since adaptive optimizers like Adam and Muon rescale

gradient magnitudes, meaning that optimization depends primarily on the correct gradient direction rather than linear proportionality. All results are summarized in Table 2.

From Table 2, we observe that wavelet-based distance functions achieve the most consistent correlation with perceptual data across all three modulation types, performing best on average for both rank order (ρ) and linear scaling (r). Scat1D matches or outperforms JTFS in all scenarios, indicating that filtering along the log-frequency axis yields no additional perceptual alignment for the listening test stimuli. Consistent with Sec. 3.1, STFT-based methods and EnCodec 48 kHz fail to capture modulation frequency, though MSS Revisited performs strongly on amplitude and irregularity. Neural representations exhibit isolated competencies rather than universal fidelity: MS-CLAP attains the highest correlation for frequency, but loses significance for irregularity, while VGGish loses significance for amplitude. Among neural methods, PANNs WGLM is the most consistent across all three modulation types. Across all evaluations, statistically significant Spearman and Pearson coefficients remain close, confirming that methods preserving perceptual monotonicity also maintain proportional distances.

4. CONCLUSION

Our listening study demonstrates that human perception of timbral modulation scales linearly with modulation amount while remaining largely invariant to underlying timbre qualities and acoustic sources. Benchmarking nine distance functions across three families (STFT-based, wavelet-based, and neural) through linear regression, variance decomposition, and correlation analyses, reveals that wavelet scattering transforms achieve the best alignment with listener ratings across modulation depth, rate, and irregularity. In contrast, STFT-based methods fail to capture modulation frequency, neural embeddings display fragmented competencies, and variance decomposition exposes that algorithmic distances invert perceptual salience of modulation amount over type. These findings support adopting wavelet-based losses over conventional spectral methods for deep inverse audio problems involving spectrotemporal modulations beyond pitch and volume. By releasing our code, stimuli, and dataset, we establish an open benchmark to guide the principled selection and design of psychoacoustically aligned distance functions and embeddings.

5. COMPLIANCE WITH ETHICAL STANDARDS

Our listening study was approved by the Queen Mary Research Ethics Committee, see the complementary website for more details.

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